Label Coverage Bounds Pooled Macro F1: A Reporting Diagnostic for Closed-Set IDS Benchmarks

Abstract We give a pre-fit check for pooled macro F1 under a declared legal output vocabulary. On the original-label NSL-KDD split, 17 of 38 test labels are absent from training and the training-restricted ceiling is 0.55263. Reaggregating the same five forest prediction matrices, including true and predicted labels, to five attack categories changes mean F1 from 0.23953 (SD 0.00878) to 0.48349 (SD 0.00549) and raises the ceiling to 1; this changes the target rather than the model. A published NSL-KDD study likewise evaluates five categories while describing test-only attack subtypes. CICIDS2017 leave-one-day-out scores 0.06030, 90.5

Macro F1, label coverage, benchmark protocols, intrusion detection, reproducibility

Introduction

A low benchmark score can describe two different failures: a model misclassifies a label it can output, or the evaluation asks it to name a label outside its output vocabulary. These failures have different remedies. Better features or fitting might address the first. They cannot address the second without changing the model's permitted decisions.

NSL-KDD provides a natural example: 21 of the 38 fine-grained labels in its official test set occur in training. A classifier restricted to training labels therefore has a positive-test-support ceiling of 21/38 (0.55263). CICIDS2017 leave-one-day-out is more extreme: its 14 attack labels are each confined to one capture day, giving a 15-class ceiling of 1/15. The attack schedule is known; this paper records the scoring consequence under an explicit label inventory.

The coverage inequality follows directly from recall and precision; we do not present it as a new theory of F1. The contribution is an operational audit for pooled scoring when legal vocabularies differ across folds: it computes the fraction of each class's test rows that can be named, makes the chosen output inventory explicit, and distinguishes training-restricted models from global-head and open-set tasks. We add a paired label-resolution analysis on NSL-KDD and map the audit to a published evaluation practice. The broader search has not established a published score that violates its legal-output ceiling, so impact on prior reported conclusions remains unproven. An optional vocabulary oracle accounts for false positives sent to legal destinations; its proofs and computational details are in the supplement.

The empirical examples illustrate how the same fixed predictions score under different label inventories. These are different tasks, so score differences are descriptive rather than evidence that one model is better. In a published NSL-KDD study, Ayantayo et al. describe 17 test-only fine-grained attack types, map the task to five broad categories, and also construct a test subset without those unseen subtypes [1]. Under that five-category output space, the five categories are

available in training and the ceiling is one; the reported category task does not establish fine-grained attribution of unseen subtypes. This is a verified example of target resolution in published practice, not an incompatible score. Our broader frozen screen still establishes no published score conflict.

Related Work

Security evaluations can become misleading when the sampling protocol answers a different question from the intended deployment setting [2], [3]. Dataset surveys and CICIDS2017 audits document traffic-generation, feature and labeling limitations [4], [5], [6]. We condition on the released labels and a declared partition; the present analysis neither repairs those datasets nor removes their other biases.

Macro F1 itself requires an explicit definition. Averaging class F1 differs from taking the harmonic mean of macro precision and recall [7]. Pooling out-of-fold predictions also differs from averaging fold scores [8]. Work on classification metrics studies prevalence and prediction bias [9], [10], while confusion-matrix objectives can guide learning algorithms [11]. Our optimization uses the true test labels to bound a fixed evaluation cohort; it is not a learning procedure.

Label shift concerns changed class priors under assumptions on conditional feature distributions [12]. A missing legal output is a separate constraint: reweighting cannot create that output. Open-set recognition and selective prediction add unknown or reject decisions [13], [14], [15], [16]. Generalized zero-shot learning permits unseen outputs through side information and evaluates seen and unseen recognition separately [17]. These settings explain why absence from training alone is insufficient to establish zero recall. Temporal and spatial discipline, as in TESSERACT [18], remains necessary even when every legal label is covered.

A Score Limit from Legal Outputs

Let f index folds, T_f their training labels, and $V_f \neq \varnothing$ their legal output vocabularies. Assume a nonempty pooled test cohort. Every test row receives one label in V_f. We allow V_f to contain labels absent from T_f; training-restricted classifiers are the special case $V_f = T_f$. Write n_{fk} for fold-level test support and define

$$S = \left\{ k : \sum_f n_{fk} > 0 \right\}, \quad n_k = \sum_f n_{fk},$$
$$m_k = \sum_f n_{fk} \mathbf{1}\{k \in V_f\}, \quad \rho_k = m_k / n_k \ (k \in S).$$

The primary metric pools confusion counts and averages F1 over positive-test-support labels S. We also report the inferred union of true and predicted labels, the default label inventory when scikit-learn's `f1_score` is called without an explicit `labels` argument. These conventions can differ when a legal output has no test support. Train-only or other output-only labels remain in the vocabulary input for both calculations.

Proposition 1. Coverage ceiling.
Every legal prediction vector satisfies

$$F \le C = \frac{1}{\mid S \mid} \sum_{k \in S} \frac{2\rho_k}{1+\rho_k}.$$

Proof. A true positive needs a legal true label, hence $TP_k \le m_k$ and recall is at most ρ_k. Precision is at most one. Substitution in the increasing harmonic-mean expression for F1 proves the result. Equality needs every available row correct and no false positives entering a positive-m_k class.

Support-weighted F1 replaces $1/\mid S \mid$ with $n_k / \sum_j n_j$. Micro F1, with all true and predicted labels included, equals accuracy and is at most $\sum_k m_k / \sum_k n_k$. Balanced accuracy is at most $\mid S \mid^{-1} \sum_k \rho_k$. These are deterministic cohort limits, not confidence intervals.

For leave-one-group-out using all remaining groups as training and $V_f = T_f$, a label in one group has $\rho_k = 0$; a label in two or more groups has $\rho_k = 1$. Thus fractional pooled coverage requires another structure, such as a forward or restricted-window protocol. This explains both the severity and the simple form of the CICIDS2017 result.

Legal outputs and limited training support

A global head permitting every test label has $C = 1$, including when some labels have no training examples. This is the correct legal ceiling. It is not evidence that the missing classes can be learned. There is no distribution-free improvement based only on training counts: a feature can encode its correct global label, and a fixed classifier can decode it perfectly without any examples of that label. Conversely, the same counts can accompany uninformative features.

For sensitivity analysis, define $V_f(\tau) = \left\{k : N_{fk}^{\text{train}} \ge \tau\right\}$. Recomputing the bound describes how coverage changes with a declared support requirement. It is a hard ceiling only for a model that enforces this restricted vocabulary. Calling it a learnability ceiling for unrestricted networks would require additional assumptions about features, priors, side information and learning. The package therefore accepts legal vocabularies explicitly and uses training labels only as the default.

Vocabulary Oracle

The coverage ceiling counts unavailable true labels but assumes all available rows are classified correctly and creates no false positives. The exact vocabulary oracle O is the maximum pooled macro F1 under perfect classification of available rows and optimal legal destinations for unavailable rows; it includes false positives created by those destinations. When exact optimization is not run, the sandwich bounds $L \le O \le U$ use the sum and the maximum of the foldwise minimum destination costs, respectively. These quantities describe attainable-score limits, not model performance; their derivation and the grouped solver are in the supplement.

Evaluation Design

We use CICIDS2017 [19], the official UNSW-NB15 partitions [20], original-label NSL-KDD [21], and the UCI gas-sensor drift archive [22]. Counts, input hashes, membership manifests and confusion matrices accompany the code. Primary results use pooled positive-test-support macro

F1 and source-training-label outputs; the NSL pipeline comparison also reports inferred-union scores. UNSW and NSL forests use 200 trees; CIC forests use 100 trees. Seeds 42–46 describe initialization variation on fixed cohorts, not independent networks or test sets.

CICIDS2017 compares five-fold stratified shuffled cross-validation with capture-day holdout. Numeric imputation uses training medians; web-label spelling variants are normalized. The local MachineLearningCVE CSVs contain no timestamps. A historical within-day row-order result is retained in the artifact but is not described as a verified temporal cutoff. All six archived forest configurations, including bootstrap weighting and that historical split, have separate fresh replication records agreeing with archived metrics to 10^{-12}.

NSL-KDD uses its named training and test files. Randomized pipelines were fit with seeds 42–46. GaussianNB and kNN are deterministic under these settings; GaussianNB was refit five times and kNN's identical deterministic predictions were reused. The forest retains its existing one-hot encoding; the five added pipelines use one-hot nominal encoding and training-fitted numeric standardization. The comparison therefore concerns complete pipelines, not isolated estimators. UNSW uses training-fitted ordinal nominal encoding. Its controlled study withholds every pair of the nine attack labels, retains all original test rows and repeats 36 pairs over five seeds. For ranking analysis we separately save confusion matrices on available and unavailable rows. Full scores retain all ten test labels; covered-only scores retain the labels with positive support in that subset. Comparisons across different pairs consequently change the covered target and are descriptive condition comparisons.

The gas study fits batch t and tests batch $t + 1$, for nine adjacent-batch folds, a protocol used in drift research [23]. The 19 fits span forests, MLPs, histogram boosting, logistic regression, SVM, kNN and CORAL logistic regression [24]. CORAL uses unlabeled target features and is transductive. It retains source labels. Ranking statistics use fixed predictions; shared rows and overlapping training conditions preclude iid significance claims.

Results

NSL-KDD: a natural partial-coverage split

Of the 38 test labels, 17 are unavailable to a training-restricted classifier. The positive-test-support ceiling and oracle are 0.55263. Two train-only labels can receive predictions without adding a class to that score's denominator. Under inferred-union scoring those destinations are scored as zero-F1 classes, and the exact oracle is 0.54837. Across five forest seeds, mean full-cohort F1 is 0.23953 (SD 0.00878); no unavailable test label is correctly predicted. The constraint is material to the attainable score, while model errors account for a large remaining gap.

The five-category mapping has ceiling one. This does not repair fine-grained recognition; it asks whether the system identifies a broader category. Reporting a high category score and a low fine-label score as interchangeable evidence would obscure this distinction.

Table 1: NSL-KDD forest scores on the same official test rows. Covered scoring removes unavailable rows and changes the scored label set.

Seed	Full support-scored F1	Covered support-scored F1
42	0.25456	0.48702
43	0.23488	0.45491
44	0.23764	0.45548
45	0.23850	0.45566
46	0.23205	0.45612

These five fits share one training file and one test file. Full scores range from 0.23205 to 0.25456; covered scores range from 0.45491 to 0.48702. We report the seed range as fit variation on this fixed cohort and do not interpret the small seed-level order changes as ranking stability or population uncertainty.

Prespecified pipeline comparison.

We compare all six prespecified pipelines on the same official NSL-KDD split. For the four randomized pipelines, table entries are means and standard deviations over seeds 42–46. GaussianNB and kNN are deterministic under these settings; each has one distinct prediction matrix, so their SD across the five seed labels is zero. GaussianNB was refit five times and kNN's identical deterministic predictions were reused. The forest and added pipelines use different preprocessing, so these results compare complete pipelines rather than estimator families. Full and covered-row scores target different label inventories; neither is a substitute for the other.

Table 2: NSL-KDD pipeline scores under positive-test-support scoring. Values are mean $\pm$ SD across five declared seeds.

Pipeline	Full F1	Covered F1
Random forest	0.23953 ± 0.00878	0.46184 ± 0.01409
Extra Trees	0.22314 ± 0.00162	0.45095 ± 0.00048
Decision tree	0.24664 ± 0.00825	0.50294 ± 0.00985
Gaussian NB	0.17968 ± 0.00000	0.34711 ± 0.00000
SGD logistic	0.20319 ± 0.01521	0.40073 ± 0.02966
kNN	0.25160 ± 0.00000	0.53995 ± 0.00000

Table 3: The same NSL-KDD predictions scored over the inferred union of true and predicted labels. Values are mean ± SD.

Pipeline	Full F1	Covered F1
Random forest	0.23338 ± 0.00855	0.44084 ± 0.01345
Extra Trees	0.21742 ± 0.00158	0.43046 ± 0.00046
Decision tree	0.24032 ± 0.00803	0.48007 ± 0.00940
Gaussian NB	0.17507 ± 0.00000	0.33133 ± 0.00000
SGD logistic	0.19588 ± 0.01274	0.37548 ± 0.02235
kNN	0.24515 ± 0.00000	0.51540 ± 0.00000

Among these six complete pipelines, kNN has the highest point estimate under both scoring conventions. On full positive-support scoring, its 0.25160 mean is 0.00496 above the decision tree's 0.24664, smaller than the tree's 0.00825 seed SD; kNN is deterministic here, so this comparison does not quantify model-to-model uncertainty. On covered rows its mean lead over the tree is 0.03701, while the covered-row task remains conditional and differs from full scoring. These results do not isolate estimator effects or establish a general performance ranking.

We also reaggregate each saved forest confusion matrix to the dataset's five broad categories without refitting. The fine-label mean remains 0.23953 (SD 0.00878) under a ceiling of 0.55263; the projected five-category mean is 0.48349 (SD 0.00549) under a ceiling of 1. The 3,750 rows carrying unavailable fine labels map to categories present in training. This paired calculation makes the role of label resolution concrete: category-level scoring accepts correct family predictions even when the exact attack subtype is unavailable. It is not a comparison between separately trained fine- and coarse-label models.

Table 4: Label resolution changes the score and ceiling for the same five saved NSL-KDD forest prediction matrices. Both true and predicted labels are mapped to the five categories.

Target	Ceiling	Macro F1, mean ± SD
Fine attack subtype	0.55263	0.23953 ± 0.00878
Five attack categories	1.00000	0.48349 ± 0.00549

CICIDS2017: a day holdout changes the scoring task

Every attack class appears on one day. Benign traffic remains available whenever tested, giving $\rho_{\mathrm{BENIGN}} = 1$; the 14 attack fractions are zero. Hence $F = F_{\mathrm{BENIGN}}/15$ and $C = O = 0.06667$. The unweighted forest's score changes from 0.85347 in shuffled cross-validation to 0.06030 in day holdout. Under the latter protocol the attack F1 terms are fixed at zero, so their contribution cannot distinguish these training-restricted classifiers.

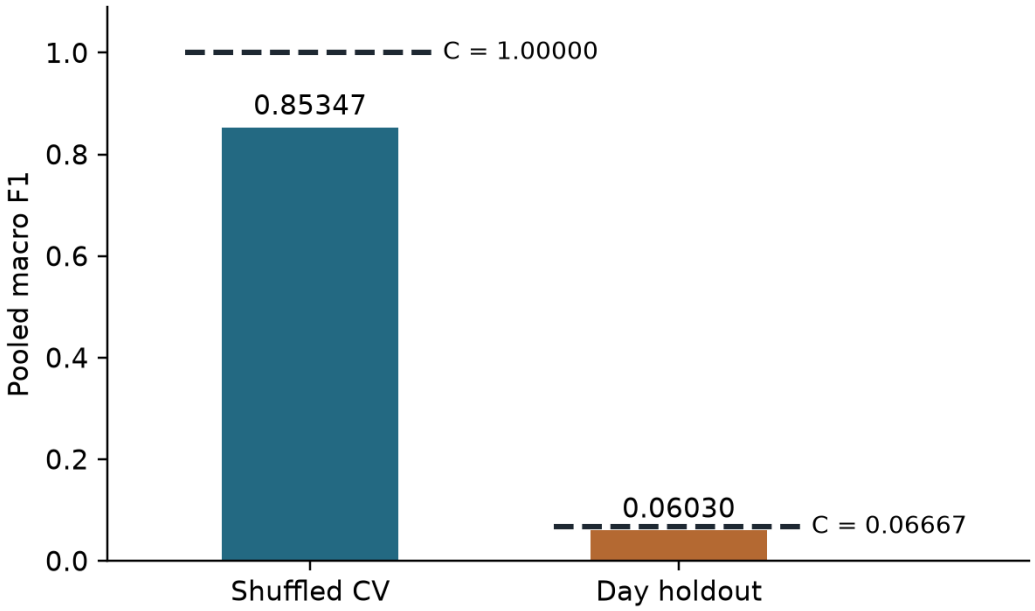


Figure 1: CICIDS2017 unweighted forest. Shuffled cross-validation has full label coverage; day holdout makes all fine-grained attack labels unavailable. The ceiling is a property of each protocol.

The day-held-out unweighted forest has benign recall 0.99858. Of the 557,646 unavailable attack rows, 475,727 are sent to benign and the remainder to other attack labels. Those other attack labels have zero covered support in the pooled score, so they provide free oracle destinations. This explains why the exact oracle equals the coverage ceiling here. The micro ceiling is 0.80300 because benign dominates the cohort. Merging web labels gives a 13-label ceiling of 0.07692; the declared eight-family mapping gives 0.25; binary attack detection has ceiling one. These are different evaluation targets.

To show what that target choice changes on the saved day-holdout predictions, we map both truth and predicted labels to the declared eight-family taxonomy and to benign-versus-attack. The same matrix scores 0.06030 at 15 fine labels, 0.15630 at eight families, and 0.57972 as a binary task; the corresponding ceilings are 0.06667, 0.25, and 1. The fine-label score is 90.5% of its ceiling because benign F1 is 0.90456 while all 14 attack-class F1 terms are zero. This attainment ratio is not a measure of attack detection. Binary attack recall is 0.14690, so collapsing labels removes the structural ceiling but does not make this fitted multiclass forest a useful detector. This post-hoc reaggregation is a task comparison on fixed predictions, not a refit or an estimate for a separately trained binary model.

Table 5: CICIDS2017 day-holdout predictions re-scored at three label resolutions. The same pooled confusion matrix is reaggregated on both axes.

Target	Classes	Ceiling	Macro F1	F/C
Fine attack labels	15	0.06667	0.06030	0.90456
Declared attack families	8	0.25000	0.15630	0.62518
Benign versus attack	2	1.00000	0.57972	0.57972

Fixed-cohort withholding and rankings

The official UNSW-NB15 split has complete coverage. The unweighted forest's five-seed mean is 0.47144, with seed SD 0.00094. Training-label availability cannot explain that shortfall. Withholding two labels creates $C = 0.8$, but its oracle depends on their support and legal destinations. For Analysis plus Backdoor at seed 42, $O = 0.79833$ and $F = 0.47067$.

The covered-row calculations use a different label inventory for each withheld pair, while full scoring retains all test labels. On this finite grid, 2,577 of 16,110 pair-seed comparisons and 97 of 630 pair-mean comparisons reverse order between the two scoring definitions. These counts are descriptive; they do not establish that either ranking is wrong. All 180 full scores match the earlier grid to absolute tolerance 10^{-12}, and the saved confusion matrices permit reconstruction of covered-row scores.

The post-hoc routing control is moved to the supplement. It uses the true availability status and illustrates how a prediction to an unscored label can affect support-scored F1; it is not evidence of improved recognition or a deployable unknown-attack rule.

Natural fractional coverage and model families

In adjacent-batch gas sensing, toluene is unavailable when batch 6 is tested. The batch-6 test fold contains 467 toluene rows; 1,759 is the pooled toluene support across all test batches. Thus 1,292 of 1,759 toluene test rows are in folds where toluene is available, giving $\rho = 1292/1759$. The other five classes have full coverage. Global label presence misses this fold-level restriction: all six labels occur somewhere in training.

Table 6: Different natural protocols retain their own targets. The NSL value is a five-seed mean $\pm$ sample SD; the other displayed model scores are seed-42 fits. $O - F$ is a score shortfall, not an estimate of model quality.

Case	C	O	F
NSL original, forest (five-seed mean)	0.55263	0.55263	0.23953 $\pm$ 0.00878
CIC day, forest	0.06667	0.06667	0.06030
Gas, forest	0.97449	0.96222	0.57034
Gas, logistic	0.97449	0.96222	0.61946

The gas oracle is 0.01227 below the coverage ceiling. This difference is 3.0% of the forest's gap $C - F$ to the ceiling and 3.5% of logistic regression's gap; the denominator is the model's shortfall from C, not from one. It explains zero toluene hits in batch 6 but little of the pooled shortfall. Across all 19 gas fits, full and covered-only rankings agree exactly: Spearman and Kendall correlations are one, with 161 concordant comparisons and ten ties. The model-family contrast therefore provides no natural family-ranking reversal. CORAL logistic scores 0.63760 while retaining the same legal ceiling.

Algorithm checks and literature screening

Exhaustive row-level enumeration on 80 small seeded inputs agrees with both solvers under support and union scoring. Another 100 cases with explicitly supplied legal heads, including labels absent from training, agree with exhaustive predictions. A repeated-vocabulary check reduces 300 unavailable folds from 2^{100} candidate assignments to two. These checks validate implementation and edge cases; fitted scores lying below a theorem are not independent evidence of significance. Varied synthetic shapes still reach exact-search budgets, as documented in the supplement.

A local protocol was committed before collecting six Crossref searches, each retaining the first 100 relevance-ranked records dated 2017–2026. Title filtering and DOI deduplication yielded 385 candidates. NumPy seed 20261108 selected 50 without replacement. The public frame, source metadata, sample, checksum and coding sheet are retained. This is a prospective local freeze rather than external preregistration.

Screening confirmed three eligible evaluations, excluded fourteen records and left thirty-three eligibility decisions unresolved. Temporal neuron coding and split learning were frequent reasons not to infer evaluation timing from a title. Of the three confirmed cases, one binary source-to-target evaluation permits both labels and therefore has legal ceiling one; its exact F1 averaging remains unresolved. The other two lack verified vocabulary/support/scoring information. No complete incompatible-score comparison is established. The result is a candidate-screening pilot, not a survey of fifty confirmed temporal IDS papers or a field prevalence estimate. Accessible abstract evidence can establish eligibility without establishing a score convention; missing evidence is retained as missing. The separate targeted source checks and DOI verification are in the supplement and artifact.

Using the Diagnostic

Choose the evaluation target before optimizing a split. Stratified shuffled cross-validation can measure interpolation over a shared inventory but does not establish future-day generalization. A chronological or capture-group holdout addresses a different deployment question and should retain the availability audit. Leave-one-attack-class-out should specify a binary alarm or a scored unknown target when the aim is unfamiliar-attack detection. Exact identification of a withheld fine-grained label cannot measure that capability for a training-restricted output head.

The reporting sequence is short: preserve fold membership; state the legal vocabulary and per-class support; declare pooling and the scored labels; compute C, then O or its sandwich if destination loss matters; report full and covered-row scores with their respective inventories.

State whether labels come from training or a global head. A ratio F/O, when $O > 0$, describes attainment on that cohort only and should accompany the absolute score. It cannot make different targets or populations comparable.

The package exposes fold-array and count-based interfaces, accepts an explicit legal vocabulary per fold, and supplies a CLI for count files. Reproduction commands and a table-to-file map accompany the release. Counts, software, manifests and checksums are public artifacts; raw traffic redistribution remains subject to dataset terms.

Limitations and Conclusion

The simple ceiling is a scoring corollary of legal-output constraints. It does not measure feature separability, rare-class learnability, calibration or deployment utility. The exact oracle adds unavoidable false-positive loss and admits a vocabulary-parameterized algorithm, but this correction is small in the natural examples considered. In the NSL-KDD complete-pipeline comparison, kNN has the highest point estimate, but its full-score margin over the decision tree is smaller than the tree's seed SD, and the pipelines use different preprocessing. These fixed-split results do not establish a general estimator ranking or a reanalysis of a published model comparison. Shared-cohort repetitions describe fit variation, not population uncertainty.

The literature screening has a bounded search frame and unresolved sources. It cannot support a field-wide prevalence claim, and no verified incompatible published score is established. The Ayantayo et al. example confirms that label aggregation and unseen subtype handling occur in a published NSL-KDD evaluation, but its five-category task is internally attainable and does not validate a fine-grained claim. The same-prediction reaggregation shows why this distinction changes reported F1, while remaining an illustration on one fixed split. Timestamped CIC files and corrected label releases are absent locally, and a file-verified CICDDoS2019 study is not supplied. A permanent DOI, an externally hosted anonymous artifact and independent reproduction also remain pending.

The practical contribution is a pre-fit audit that records the task the score actually measures: the label resolution, legal outputs, scoring convention, and resulting ceiling. On NSL-KDD, changing from subtype labels to five categories changes both the bound and the same-prediction macro F1. On CICIDS2017 day holdout, binary reaggregation removes the fine-label ceiling but the existing model recalls only 14.69% of attacks; the gas case shows that a ceiling can also explain little. These cases demonstrate what the audit can and cannot resolve. They do not establish that published scores are generally invalid.

Data availability

Dataset sources are cited above. The accompanying code release contains aggregate results, split manifests, hashes, source coding and commands mapped to the manuscript. Original data terms continue to apply.

Declarations

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Use of generative AI

Generative AI assisted code development, mathematical exposition, literature screening, analysis planning and manuscript editing. Numerical results were computed with the released scripts. The author retains responsibility for the submitted work.

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